Colligative Properties: Freezing Point Depression

Chem 151 Lab

Freezing Point Depression and Boiling Point Elevation

For both freezing point depression and boiling point elevation, you use the same formula.

$$\Delta T = K_{f/b} \cdot i \cdot m$$

- i: number of particles, per molecule
- m: molality
- $K_{f/b}$: freezing dep. or boiling elev. constant

water has a $K_f = 1.858\ ^\circ C \cdot kg/mol$ and $K_b = 0.512\ ^\circ C \cdot kg/mol$

Freezing Point Depression in Solute + Water

$\Delta T = K_f \cdot i \cdot m$

$\Delta T = T_{solvent} - T_{solution}$ (should be negative because temp is depressed!)

K_f = Freezing point constant; for H_2O = 1.858 ˚C·kg/mol

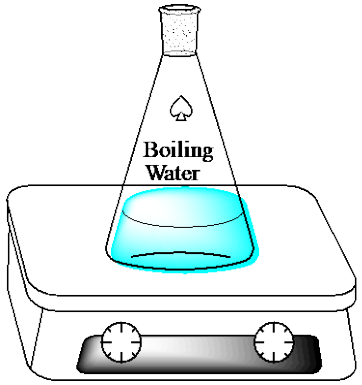
m = molality = $\frac{moles\ of\ solute}{kg\ of\ solvent}$

i = van't Hoff factor

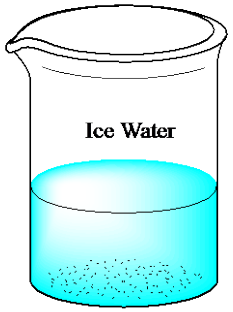
Determining this experimentally today!

Experimental Set-Up

A. Calibrate the temperature probe using two known temperatures



94.5 °C

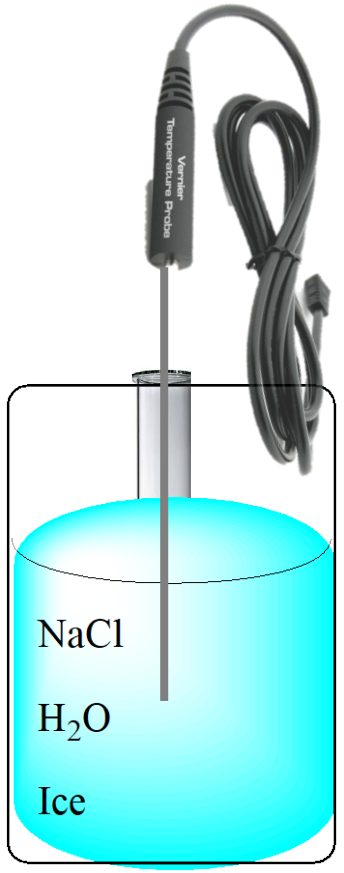


0.0 °C

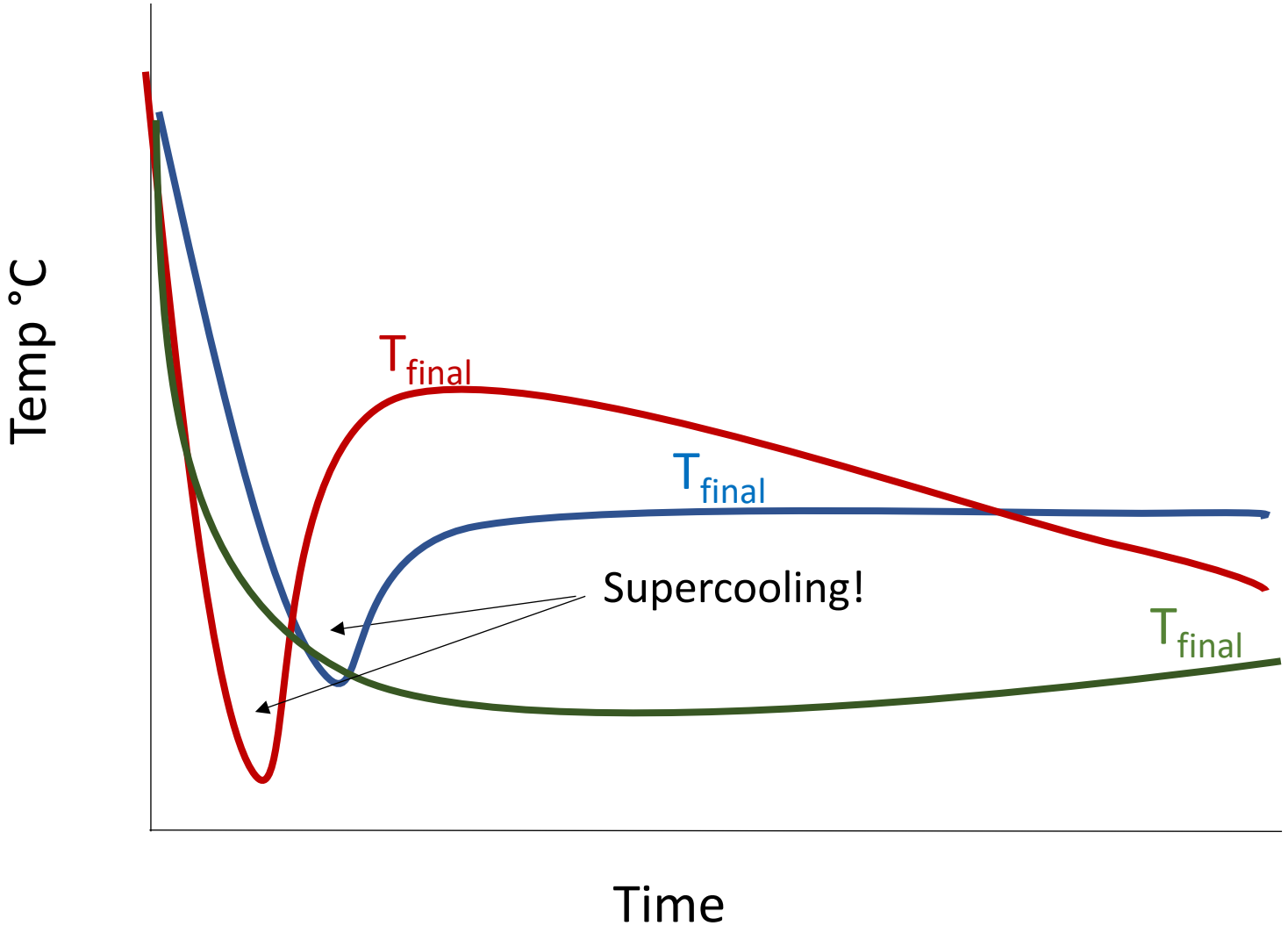
B. Use the temperature probe with three runs:

1. Pure H_2O – gives you $T_{solvent}$
2. Sucrose + H_2O
3. NaCl + H_2O

Scratch the probe along the sides of the test tube during data collection



Cooling Curves



*Caution: NaCl can supercool for a long time! Be patient!!

Announcements

- Your Colligative Properties Lab is due by electronic notebook upload to Canvas by Wednesday at 11:59PM
- Because we are working with sugar, salt and water today, you do not need to wear safety glasses/goggles and all solutions can be poured down the drain